



VCS VERIFICATION REPORT

GUANGDONG NAN AO 26MW WIND
POWER PROJECT

(VCS PROJECT ID: 254)



Document Prepared By

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Summary:

Shenzhen CTI International Certification Co., Ltd (CTI) has performed the verification of the emission reductions reported for the project activity “Guangdong Nan Ao 26MW Wind Power Project” (VCS Project ID:254) for the monitoring period 29/09/2012-31/12/2017 within the 1st VCS crediting period 01/01/2008-31/12/2017, to review and determine the monitored reductions in GHG emissions that have occurred as a result of the project activity. These emission reductions are claimed as Verified Carbon Units (VCU) under the VCS Standard Version 4.3.

The verification was performed on the basis of VCS Standard Version 4.3 for the VCS projects, as well as criteria given to provide for consistent project operations, monitoring and reporting. The verification was conducted by means of document review, follow-up interviews and site inspections, and the resolution of outstanding issues. The verification team identified one CL, and no CAR or FAR is raised in this monitoring period.

In CTI's opinion, the GHG emission reductions reported for the project in the monitoring report (version 03 dated 24/02/2023) are fairly stated. The GHG emission reductions of the period from 29/09/2012-31/12/2017 within the 1st VCS crediting period were calculated correctly on the basis of approved methodology ACM0002 “Consolidated methodology for grid-connected electricity

generation from renewable sources” (version 06), the registered CDM-PDD (version 4, dated 12/03/2008) and the revised monitoring plan approved by CDM EB on 09/07/2010.

CTI does not assume any responsibility towards the issuance and utilization of the VCUs hereby verified and certified. Request for issuance of VCUs shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to CTI and the engagement conditions detailed in this report. CTI cannot be held liable by any party for decisions made or not made based on this report.

Hence, CTI is able to certify that the emission reductions from the “Guangdong Nan Ao 26MW Wind Power Project” during this monitoring period from 29/09/2012-31/12/2017 amount to 275,489 tCO₂e.

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1 INTRODUCTION

Timing Carbon Asset Management Co., Ltd. has commissioned Shenzhen CTI International Certification Co., Ltd (CTI) to carry out the verification and certification of emission reductions reported for the “Guangdong Nan Ao 26MW Wind Power Project” (the Project) for the monitoring period 29/09/2012-31/12/2017. This report contains the findings from the verification and includes a verification statement for the verified carbon units.

1.1 Objective

Verification is the periodic independent review and ex-post determination by an accredited verification body of the monitored reductions in GHG emissions that have occurred as a result of the registered VCS project activity during a defined verification period.

A verification statement is the written assurance by a verification body that, during a specific period in time, a project activity achieved the emission reductions as verified.

The objective of this verification was to verify and provide a verification statement of emission reductions reported for the “Guangdong Nan Ao 26MW Wind Power Project” for the period 29/09/2012-31/12/2017.

1.2 Scope and Criteria

The scope of the verification is:

- To verify that actual monitoring systems and procedures are in compliance with the monitoring systems and procedures described in the approved revised monitoring plan;
- To evaluate the GHG emission reduction data and express a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement;
- To verify that reported GHG emissions data is sufficiently supported by evidence.

The criteria of the verification are:

- VCS Program Guide (version 4.2)/41/
- VCS Standard (version 4.3)/40/ and other relevant requirements defined by Verra;
- The approved methodology ACM0002 (version 06)/44/ applied by the project.

The verification shall ensure that reported emission reductions are complete and accurate in order to be verified.

1.3 Level of Assurance

The verification report expresses a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement. As per the para 3.9.1 of the VCS Standard Version 4.3, the project size categorization belongs to 'Project' as the expected annual emission reductions is 53,425 tCO₂e lower than 300,000 tCO₂e. Therefore, according to the requirement of para 4.1.8 (4) of VCS standard Version 4.3 "The threshold for materiality with respect to the aggregate of errors, omissions and misrepresentations relative to the total reported GHG emission reductions and/or removals shall be five percent for projects and one percent for large projects.", CTI applied a materiality threshold of 5% with respect to omission or misstatements concerning reported quantities of the project activity.

1.4 Summary Description of the Project

Sectoral Scope and Project Type

According to the VCS Program Guide (version 4.2)/41/ and Annex A of the Kyoto Protocol, the project is applicable under the following activity categories:

- Sectoral Scope: 1. Energy (Renewable/non-renewable)
- Project Type: Grid-connected wind power project

Project Background

The project is located on Nan Ao Island, Nan Ao County, Shantou City, Guangdong Province P.R.China. The site location's approximate coordinates are east longitude of 117°5'27.73" and north latitude of 23°25'12.63". The project activity was registered as a CDM project with Ref.1963 on 15/12/2008 and the renewable crediting period was selected starting on 15/12/2008. The project has been registered as VCS project with VCS ID. 254 under VCS Standard Version 2007.1 with the crediting period start date 01/01/2008 when the commencement of operations activities started as confirmed by checking the Commissioning Report /29/, operation logs/7/, the VCS monitoring report and verification report for the period of 01/01/2008-14/12/2008 /24//25/.

The amount of 48,524 tCO₂e VCUs have been issued for the monitoring period 01/01/2008-14/12/2008 within the 1st VCS crediting period. GHG emission reductions have been applied for CERS issuance under CDM of which total 193,689 tCO₂e have been issued for the monitoring periods: 15/12/2008-28/10/2009, 29/10/2009-28/09/2010, 29/09/2010-28/10/2011 and 29/10/2011-28/09/2012.

The installed capacity of the project is 23.25MW equipped with 31 sets of wind turbine generator in type of S48/750, with unit capacity of 0.75MW manufactured by Goldwind Science & Technology Co., Ltd. The electricity generated by the project activity is supplied to the China Southern Power Grid (CSPG), and the project is estimated to deliver 53,425 tCO₂e emission reductions annually.

2 VERIFICATION PROCESS

2.1 Method and Criteria

The verification was performed through means of the following three phases in accordance with the requirement of the registered PDD (version 4 dated 12/03/2008) and the revised monitoring plan approved by CDM EB on 09/07/2010, the applied methodology, and the VCS Standard (Version 4.3) and other relevant VCS requirements:

- A desk review of the monitoring report and all support documents;
- Follow-up interviews with project stakeholders and site inspection;
- The resolution of outstanding issues and the issuance of the verification report and statement.

The following sections outline each step in more detail.

The verification of the emission reductions has assessed all factors and issues that constitute the basis for emission reductions from the project. These include:

- The emission reduction calculations and the relevant data records;
- The calibration and maintenance records for the monitoring instruments;
- The management systems to support the project operation and monitoring.

2.2 Document Review

Based on the requirements of competency, experience and qualified sectoral scopes, CTI appointed a verification team in accordance with CTI's internal procedures.

Function	Name	Technical competence	Task Performance*
Team Leader& Verifier	Feng Tian	1.2	☒ DR ☒ SV ☒ RP ☐ TR
Technical Reviewer	Lin Shunrong	1.2, 14.1, 15.1	☒ DR ☐ SV ☐ RP ☒ TR

*DR=Document review; SV=Site visit; RP=Reporting; TR=Technical review

In addition to the VER/VCU monitoring report/1/, the registered PDD /10/, emission reduction calculation spreadsheet/2/ during this monitoring period and revised monitoring plan approved by CDM EB on 09/07/2010, the following documents also were assessed as a part of the verification audit:

- The CDM registered PDD and corresponding validation report /10/;
- The previous VER/VCU monitoring report/24/ and verification report/25/
- The previous CDM/CER monitoring reports/26/ and verification reports/27/;
- Baseline and monitoring methodology ACM0002 (version 06) applied by the project/44/;

- Relevant decisions, clarifications and guidance from the Verra/40/-/43/; and
- Other information and references relevant to the project activity.

During the desk review, CTI has applied standard auditing techniques to assess the quality of information provided. The following activities were performed:

- A review of the data and information presented to verify their completeness;
- A review of the approved revised monitoring plan and monitoring methodology, paying particular attention to the frequency of measurements, the quality of metering equipment including calibration requirements, and the quality assurance and quality control procedures; and
- An evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions.

2.3 Interviews

On 01/08/2022, CTI visited the project proponent China Resources Wind Power (Shantou) Co., Ltd. to perform on-site assessment. The key personnel of the project were interviewed or assisted the verification team /45/. Main topics of the interview cover implementation of the project construction, applicability of selected methodology, implementation of project monitoring, emission reduction calculation, etc.

The key personnel interviewed /45/ are summarized in the table below:

Interviewed personnel	Role	Organization	Subject
Mr. Xiao Weinan	Plant Director	China Resources Wind Power (Shantou) Co., Ltd.	Operation of the project activity; Implementation of the monitor plan of the project activity; Data collection and data achievement; Calibration of meters and equipment maintenance; The complaint by local stakeholders; The stakeholder consultation during the operation of the project activity. Project's sustainable development contributions.
Mr. Ke Dehao	O&M Staff		
Mr. Zhang Donf			
Mr. Hou Xiaohe			
Mr. Chen Ruibiao	Local resident	Local Village	The impact of the project activity; The complaint by local stakeholders; The stakeholder consultation during the operation of the project activity; Project's sustainable development contributions.
Mr. Ke Qiushan			
Mr. Ke Shusong			
Mr. Lin Zeqin	Staff	Local Environment Protection Bureau	
Ms. Liu Yanyan	Project Manager	Timing Carbon Asset Management Co., Ltd.	Data collection and ER calculation.

2.4 Site Inspections

During the on-site assessment, CTI has applied standard auditing techniques to assess the quality of information provided. The following aspects of the project activity have been verified:

- An assessment of the implementation and operation of the registered project activity is as per the registered PDD of the project activity;
- A review of information flows for generating, recording, aggregating and reporting the monitoring parameters; and
- Interviews with relevant personnel to determine whether the operational and data collection procedures are implemented in accordance with the approved revised monitoring plan;
- A cross-check between information provided in the monitoring report and data from other sources such as plant logbooks and the electricity transaction notes (ETNs);
- A check of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the approved revised monitoring plan and the selected methodology;
- A review of calculations and assumptions made in determining the GHG data and emission reductions; and
- An identification that quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

In addition, all parameters required by the monitoring methodology ACM0002 (version 06), and the management system were assessed during the site visit.

2.5 Resolution of Findings

A corrective action request (CAR) shall be raised, where:

- i. Non-conformities with the monitoring plan or methodology are found in monitoring and reporting, or if the evidence provided to prove conformity is insufficient;
- ii. Mistakes have been made in applying assumptions, data or calculations of emission reductions which will impair the estimate of emission reductions;
- iii. Issues identified in a FAR during validation to be verified during verification have not been resolved by the project proponents.

A clarification request (CL) shall be raised if information is insufficient or not clear enough to determine whether the applicable VCS requirements have been met.

The verification team identified 1 CL and no CAR in this monitoring period. The CL was satisfactorily addressed by the project proponents in the updated monitoring report version 03 dated 24/02/2023 (refer to Appendix D for further details).

2.5.1 Forward Action Requests

A forward action request (FAR) is issued for actions if the monitoring and reporting require attention and/or adjustment for the next monitoring period.

CTI confirmed that there was no FAR identified in previous verification/25/, and no FAR was raised during this verification.

2.6 Eligibility for Validation Activities

The project has been validated by TÜV SÜD Industrie Service GmbH under the CDM rules and registered as CDM project on 15/12/2008, and was registered as VCS project under VCS standard version 2007.1 through gap validation by TÜV NORD CERT GmbH.

CTI was accredited in Sectoral Scope 1 on validation and verification and thus is eligible for verification activities on the project. By checking the UNFCCC website, the VCS pipeline and the previous verification report, it is confirmed the verification for this monitoring period 29/09/2012-31/12/2017 which is no more than consecutive 6 years is the first verification undertaken by CTI. Therefore, the verification undertaken by CTI is in line with the requirement for VVB stipulated in Section 4.1.20 of VCS standard Version 4.3

3 VALIDATION FINDINGS

3.1 Participation under Other GHG Programs

The project has been registered as a CDM project on 15/12/2008 under the UNFCCC with Ref. 1963. The renewable CDM crediting period was selected starting on 15/12/2008. By checking the UNFCCC website: <https://cdm.unfccc.int/Projects/DB/TUEV-SUED1217248156.51/view>, it is confirmed that total 193,689 tCO_{2e} of CERs have been issued for the monitoring periods: 15/12/2008-28/10/2009, 29/10/2009-28/09/2010, 29/09/2010-28/10/2011 and 29/10/2011-28/09/2012.

3.2 Methodology Deviations

The validation process /16/ has assessed all factors and issues that constitute the basis for emission reductions from the project according to the applicable CDM methodology ACM0002 (version 06) /44/. There was not any methodology deviation applied to this project. Details refer to section 4.1.

3.3 Project Description Deviations

By checking the previous VCS verification reports with the monitoring period 01/01/2008-14/12/2008, the Commissioning report of the project /29/, operation logs/7/ and through the site visit interview with

project owner, it is confirmed that the project started operation on 01/01/2008 which is the project start date according to the requirement of VCS. A deviation was identified for the crediting period of the Project. The project is registered as VCS project under VCS version 2007.1 and completed validation before 19/03/2020. As per VCS requirement, it remains eligible to apply the crediting period requirements under VCS Version 3 which shall be a maximum of ten years and may be renewed at most twice. Therefore, the first renewable crediting period of the Project under VCS is updated from 01/01/2008-14/12/2008 to 01/01/2008-31/12/2017, which lasts for 10 years. As per VCS requirement, the project did not conduct the validation of crediting period renewal within the two years after the end of the first VCS crediting period. Thus, the crediting period of the project ends on 31/12/2017.

CTI confirms that the project remains in compliance with VCS Program rules and the deviation on the crediting period has no impact on the applicability of the methodology, additionality of project activity, and the appropriateness of the baseline scenario.

By checking the previous verification reports and the UNFCCC website: <https://cdm.unfccc.int/Projects/DB/TUEV-SUED1217248156.51/view>, a revised monitoring plan has been approved by CDM EB on 09/07/2010 due to that the Project shares the same meters (M1 and M3) installed at the 110kV local substation with the other wind farm, namely Nanguo Wind Farm which was put into operation since 11/2008. The revised monitoring plan specifies the monitoring and calculations of net electricity supplied to the grid by the project activity (EG_y) as follows:

$$EG_y = EG_{\text{output},y} - EG_{\text{Nanguooutput},y} - EG_{\text{import},y}$$

Where:

EG_y is the net electricity supplied by the project,

$EG_{\text{output},y}$ is the electricity supplied by the proposed project and the Nanguo wind farm;

$EG_{\text{Nanguooutput},y}$ = Electricity generated by Nanguo wind farm

$EG_{\text{import},y}$ is the electricity purchased from the Grid by the project and the Nanguo wind farm.

Of which, $EG_{\text{Nanguooutput},y} = \sum_{i=1}^8 EG_{M2,i}$

Where,

$EG_{M2,i,y}$ is the electricity supplied by turbine i of Nanguo wind farm measured by Meter M2.i;

i = The number of turbine of Nanguo wind farm, from 1 to 8.

The verification team confirmed that the deviations did not lower the accuracy and conservativeness of the monitoring system and will not negatively impact the conservativeness of the quantification of GHG emission reductions or removals. Furthermore, the deviation relates only to the criteria and procedures for monitoring or measurement, doesn't relate to any other part of the methodology.

In conclusion, the verification team assessed through visual inspection and document review that all physical features of the project activity including wind turbines, generators, transmission facilities, monitoring equipment, data monitoring, reporting and collecting systems have been implemented in accordance with the registered PDD and approved revised monitoring plan. CTI confirms that the project is implemented in accordance with the project description, and no any other project description deviations occurred. Details refer to section 4.1.

3.4 Grouped Project

The project was not a grouped project; hence this clause is not applicable.

4 VERIFICATION FINDINGS

This section summarizes the findings from the verification of the emission reductions reported for the “Guangdong Nan Ao 26MW Wind Power Project” for the monitoring period 29/09/2012-31/12/2017.

4.1 Project Implementation Status

Project Implementation in accordance with the registered project design document

The project is a wind power grid-connected plant and locates at Nan Ao Island, Nan Ao County, Shantou City, Guangdong Province P.R.China. The geographical coordinates are verified by GPS system as east longitude of 117°5'27.73" and north latitude of 23°25'12.63". By checking the FSR /30/, equipment purchase contract/32/ and through on-site inspection, it is confirmed that the total installed capacity is 23.25MW equipped with 31 sets of wind turbine generator in type of S48/750, with unit capacity of 0.75MW. The location and geodetic coordinates of all of the 31 wind turbines are verified by CTI through checking the GPS measurements /39/ against the Layout of the wind turbines distribution /38/ as well as the Map of the project site /37/. The coordinates of all wind turbines as summarized in the table below:

No.	Type	Longitude (E)	Latitude (N)
1.	S48/750	117°05'17.33"	23°25'53.03"
2.	S48/750	117°05'22.73"	23°25'51.15"
3.	S48/750	117°05'30.16"	23°25'52.33"
4.	S48/750	117°05'33.37"	23°25'50.31"
5.	S48/750	117°05'38.42"	23°25'46.64"
6.	S48/750	117°05'44.08"	23°25'43.10"

7.	S48/750	117°05'44.62"	23°25'38.53"
8.	S48/750	117°04'59.54"	23°25'45.17"
9.	S48/750	117°05'07.80"	23°25'28.97"
10.	S48/750	117°05'10.77"	23°25'36.39"
11.	S48/750	117°05'11.00"	23°25'40.07"
12.	S48/750	117°05'17.58"	23°25'41.33"
13.	S48/750	117°05'18.52"	23°25'34.77"
14.	S48/750	117°05'20.99"	23°25'30.56"
15.	S48/750	117°05'26.98"	23°25'32.11"
16.	S48/750	117°05'27.94"	23°25'28.46"
17.	S48/750	117°05'33.06"	23°25'28.45"
18.	S48/750	117°05'18.85"	23°25'07.72"
19.	S48/750	117°05'17.10"	23°25'04.16"
20.	S48/750	117°05'27.73"	23°25'10.06"
21.	S48/750	117°05'29.31"	23°25'07.82"
22.	S48/750	117°05'33.41"	23°25'07.95"
23.	S48/750	117°05'35.09"	23°25'01.21"
24.	S48/750	117°04'53.94"	23°25'49.08"
25.	S48/750	117°04'20.53"	23°26'16.05"
26.	S48/750	117°04'23.10"	23°26'13.16"
27.	S48/750	117°05'14.49"	23°25'28.57"
28.	S48/750	117°05'13.87"	23°25'22.00"
29.	S48/750	117°05'16.00"	23°25'16.84"
30.	S48/750	117°05'17.99"	23°25'14.11"
31.	S48/750	117°05'24.93"	23°25'15.00"

It was confirmed by checking the Commissioning report /29/, the operation logs/7/, the previous VCS monitoring report and verification report /24/-/25/ that the project was put into operation since 01/01/2008. The actual implementation of the project during this verification period was verified in terms of nameplates of wind turbines /28/ and the equipment purchase contract /32/. The details of the turbines with respect to their installation and capacity have been verified to be consistent with description indicated in the registered PDD.

The electricity generated by the project activity is supplied to the China Southern Power Grid, which can be confirmed by the Power purchase agreement (PPA) signed between China Resources Wind Power (Shantou) Co., Ltd. and Guangdong Power Grid Co., Ltd. Shantou Power Supply Bureau/3/. All the monitoring system in operation period is consistent with the description in the monitoring plan. The control system at the power plant is automated and assures continuous operation, including monitoring on malfunction of equipment. By checking the daily operation and maintenance records/7/ and via on-site visit, CTI can confirm that no serious malfunction happened and the plant was under a normal operation as expected in this monitoring period.

On-site training for the related procedures including, recording and reporting was verified to be in place/6/ and their implementation was confirmed by interview with the key operators/45/ and observing the operation.

CTI confirms that the project implementation is in accordance with the project description contained in registered PDD. Furthermore, through visual inspection and document review, it is confirmed that all physical features of the project activity including data collection systems and storage systems have been implemented in accordance with the approved revised monitoring plan.

Participation or any rejection under any other GHG programs, any other form of environmental credit or emissions trading program since validation or previous verification

By checking the UNFCCC website: <https://cdm.unfccc.int/Projects/DB/TUEV-SUED1217248156.51/view>, it is confirmed that the emission reductions within the monitoring periods 15/12/2008-28/10/2009, 29/10/2009-28/09/2010, 29/09/2010-28/10/2011 and 29/10/2011-28/09/2012 have been issued as CERs and no other issuance request was submitted under CDM.

Therefore, it is concluded no double counting issuance has occurred for this verified monitoring period 29/09/2012-31/12/2017 under both VCS and CDM programmes. Via the on-site interview, the project proponent confirmed the emission reductions for this verified period are only requested for VCU issuance.

The verification team has also checked other GHGs programs and forms of credit such as exchange platform of Chinese Certified Emission Reduction (CCER) projects, Gold Standard, EU ETS, IREC, China ETS and based on the interview with the project proponent, during this monitoring period, it is confirmed that except CDM and VCS scheme:

- No evidence showing that the project has participated or been rejected under any other GHG programs since validation or previous verification.

- There is no observation that the project has received or sought any other emission trading program and form of environmental credit, or has become eligible to do so since validation or previous verification.

The project is completely voluntary CDM and VCS project, not included in an emissions trading program or any other mechanism that includes GHG allowance trading. Although the national emission trading scheme has been launched since 16/07/2021, it only covers the high-emission and energy intensive industries that emit at least 26,000 tonnes of CO₂e/year as per the regulation/19/ including thermal power generation, petrochemical, chemical, building materials, iron and steel, etc. By checking the business license/13/of the project proponent, it is confirmed that the PP is qualified in business scope of renewable energy investment, thus is not included the mandatory emission control scheme and there is no emission cap enforced for the PP by further checking the compulsive company list under China ETS in public information/20/.

Moreover, it is confirmed by CTI the project is not registered as a CCER project in China, the emission reductions generated by the project activity are not eligible for transaction under China ETS as per the regulation/19/. Also, CCER mechanism has been suspended by Chinese government since 2017/20/, the project activity will not and is not allowed to register under CCER mechanism. Thus, the assessment team concluded that the project activity not involved on other emissions trading programs and other binding limits.

The Statement on no double counting made by PP /33/ submitted by PP was checked by the verification team and confirmed to be correct in terms of the regulations and database of CCER and China ETS.

SD contributions resulted by implementation of the project activity

Via the on-site inspection, and through the interviews with the project proponent, local villagers and official from local environment bureau, and by checking the evidences provided by the project proponent, the verification team agrees on the following points regarding to the promotion of sustainability development by the project activity:

- *SDG 7, Indicator 7.2.1 Renewable energy share in the total final energy consumption*

The project provides clean and renewable energy through making use of wind power, displacing fossil fuel power in the grid China Southern Power Grid. During this monitoring period, 296,994.687MWh of net electricity generated by wind energy was delivered to the China Southern Power Grid. It is verified by checking the daily and monthly reading records/8/ and via on-site inspection. This contributes to increasing the share of clean energy consumption of China's "Action Plan for Carbon Dioxide Peaking Before 2030". /35/

- *SDG 8, Indicator 8.5.1 Average hourly earnings of employees, by sex, age, occupation and persons with disabilities*

It is verified by interviewing with project proponent, staff and local villagers and checking the Statement on the annual income of employees from 2012 to 2017 /21/ and Per capita disposable income of residents in Guangdong Province in 2012 and 2017/22/, that the project activity provides 11-12 long-term job opportunities for local residents including 3 positions for females from 2012-2017 and provides higher salary from 72,000 RMB/year in 2012 to 123,480 RMB/year in 2017

which are 138.2% to 201.4% higher than the local average level of 30,226 RMB/year in 2012 and 40,975 RMB/year in 2017. Hence, the contribution to this SDG has been confirmed and the provisions for monitoring and reporting this goal is verified as reasonable. This also contributes to one of the China's actions for promoting sustainable developing, "Increase labor force participation rate through implementation of the classification policy. Vigorously enforce the Law on Promotion of Employment" /34/

- **SDG 13, Indicator Tonnes of greenhouse gas emissions avoided or removed**

As verified by on-site inspection, and via checking the emission reduction calculation sheet/2/ and the monthly reading records/8/, 275,489 tCO₂e GHG emission reductions are generated by the project activity during this monitoring period. This contributes achieve one of China's stated sustainable development priorities "Actively adapt to climate change and strengthen resistance capacity to climate risks in agriculture, forestry, water resources and other key fields, as well as cities, coastal regions and ecologically vulnerable areas". /36/

Meanwhile, as this is the first monitoring period that the project reported its SDG contribution, all SDG contributions during the previous monitoring periods have not been calculated in the cumulative impact. Therefore, the cumulative contributions over the project lifetime are considered as the same as the current project contributions.

In conclusion, Verification Team was able to confirm that the project implementation is in accordance with the project description contained in the registered PDD. All required equipments and procedures are available and implemented in an appropriate manner. All necessary monitoring instruments are installed and operated. The project is completely operational and the same has been confirmed on-site. Neither mistakes nor malfunction on main meters have been observed during this monitoring period.

4.2 Safeguards

4.2.1 No Net Harm

Via on-site inspection and checking the EIA summary and conclusion provided in the approved PDD, it is confirmed that the impact caused by the project activity on the surrounding ecosystem and residents, i.e. impact from noise, electromagnetic impact and impact on water quality etc. is insignificant, there would be no net harm caused due to the project activity. Appropriate mitigation measures have been undertaken by the project proponent to avoid the potential impacts during the construction and operation. The EIA of the project are approved by the government.

Also, no potential environment or social economic matter was found during the site visit. It is verified by interviewing with project proponent, staff and local villagers and checking the Statement on the annual income of employees from 2012 to 2017/21/ and Per capita disposable income of residents in Guangdong Province in 2012 and 2017 /22/, that the project has contributed to the local employment and improved income levels by providing job opportunities and offering permanent employees wages well above local average income levels. As stated by the plant staff during the on-site interviews, they have gained knowledge and skills in the regular training /6/ organized by the project proponent, so it

can be judged that the project has, to some extent, improved the marketability of the employees as well as the possibility of lifelong learning for them.

In conclusion, the implementation of the project activity does not cause any significant negative environmental or socio-economic impacts and is beneficial to the local socio-economic environment.

4.2.2 Local Stakeholder Consultation

The local stakeholder's survey was carried out in April 2007 by the project proponent in Nan Ao County. Total 50 questionnaires were given out and 48 of them were returned successfully. The questionnaire survey candidates include governmental officials of local county and local residents such as farmers, workers and students.

The questionnaire survey showed that the proposed project receives strong support from the local community. They all believe the proposed project will promote local economic development and agree with the project development and construction.

For continuous communication with local stakeholders, through on-site inspection and interview during on-site visit, it is confirmed that the project owner has published the phone number of the contact person to local people and village committees, anyone who has concerns can call the phone number to express their opinions directly. Also, the contact person meets the local residents at least once a year to collect their comments and suggestions i.e. in October 2012, November 2013, October 2014, November 2015, November 2016, and October 2017 during this monitoring period as confirmed by checking the Meeting minutes /14/ between contact person and local stakeholders. Through interviewing with contact person, staff from local environmental protection bureau and local residents during the site visit, it is confirmed by the verification team that during the implementation stage of this monitoring period, local authority has conducted spot checks on the implementation of the project periodically as per the request from the local governments' regulations, no negative comments nor grievance were received from the local stakeholders during this monitoring period were identified and the project passed all the periodic spot checks by local government.

All such conclusion has been verified through site visit and checking PDD.

4.3 AFOLU-Specific Safeguards

For non-AFOLU projects, this section is not required.

4.4 Accuracy of GHG Emission Reduction and Removal Calculations

Compliance of monitoring plan with monitoring methodology

CTI is able to confirm that the approved revised monitoring plan is in accordance with the approved methodology applied by the project activity, i.e. ACM0002 (version 06).

Compliance of monitoring with the monitoring plan

CTI confirms that the parameters stated in the monitoring plan are monitored and reported appropriately. Parameters required to be monitored by the approved revised monitoring plan as per the monitoring methodology ACM0002 (version 06) and the management system was assessed during the site visit. The monitoring report lists each parameter required by the approved revised monitoring plan and the information flow (i.e. from data generation, recording, aggregation, calculation and reporting) for each parameter is provided. The information flow for each parameter is further verified in the following sections.

Parameters fixed ex ante

“Data and parameters fixed ex-ante” in the MR are checked against the registered PDD. The parameter EF_y is the combined emission factor of the grid which was determined ex-ante at the validation stage and has been fixed during the first crediting period. CTI verified and confirms that the emission factor used in the monitoring report is in compliance with the registered PDD. Other ex-ante parameters (i.e. EF_{OM} , EF_{BM} , $F_{i,j,y}$, $NCV_{i,y}$, $EF_{CO2,i,y}$, Carbon Oxidation Factor, Installed Capacity, Electricity Generation and Electricity self-consumption ratio) were used to calculate the parameter EF_y in the registered PDD, which were validated in the validation process. CTI verified and confirms that the emission factor used in the monitoring report is in compliance with the registered PDD.

Parameters monitored

CTI conducted document review and performed on-site assessment with project stakeholders to:

- A review of information flows for generating, aggregating and reporting the monitoring parameters;
- Determine whether the operational and data collection procedures are implemented in accordance with the registered monitoring plan;
- A cross-check between information provided in the monitoring report and data from other sources such as plant logbooks, sales receipts and electricity confirmation;
- An identification that quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

According to the approved revised monitoring plan, the parameters to be monitored are as below:

$EG_{output,y}$ Electricity supplied by the Project and the Nanguo wind farm

$EG_{import,y}$ Electricity purchased from the Grid by the Project and the Nanguo wind farm

$EG_{Nanguooutput,y}$ Electricity generated by Nanguo wind farm

Monitored parameter $EG_{output,y}$, $EG_{import,y}$ and $EG_{Nanguooutput,y}$

By site inspection and checking the approved revised monitoring plan and previous verification reports, CTI confirmed that the meters involved in the project are :

Bidirectional main meter M1 (accuracy class of 0.5S) installed at the outlet of 110kV local substation and its backup meter M3 (accuracy class of 0.5S) installed at the inlet of the 110kV local substation are used to measure $EG_{output,y}$ and $EG_{import,y}$, the ETNs are issued by the grid company based on the reading records of main meter M1.

The unidirectional meters M2,1-M2,8 (all with accuracy class of 0.2S) and their backup meters M4,1-M4,8 (all with accuracy class of 0.5S) installed at the eight turbines of Nanguo wind farm are used to monitor and $EG_{Nanguooutput,y}$.

The verification team had on-site checked the location of the meters against the diagram of power connection system /4/ and found them to be consistent and installed in accordance with the approved revised monitoring plan. All the monitoring facilities and system have been verified by CTI during on-site visit.

By checking operation log /7/ and via on-site visit, it is confirmed that all meters were working well during this monitoring period. Via the on-site interview and by checking the daily operation and maintenance records/7/, monthly reading records/8/ and ETNs/9/ issued by grid company, it is confirmed by the verification team that $EG_{output,y}$ and $EG_{import,y}$ are continuously measured by M1 and M3 and monthly recorded by PP and grid company, after confirmed by both sides, the grid company issues the ETNs to the project proponent every month and the ETNs have been used for cross-checking the reading records of $EG_{output,y}$ and $EG_{import,y}$.

$EG_{Nanguooutput,y}$ is continuously measured by M2,1-M2,8 and M4,1-M4,8 and monthly recorded by PP. The cut-off time of all electricity data is at 24:00 on one day from 24th to the last day of each month which is determined by the grid company.

According to the approved revised monitoring plan, net electricity supplied to the grid by the project activity (EG_y) is calculated as follows:

$$EG_y = EG_{output,y} - EG_{Nanguooutput,y} - EG_{import,y}$$

Where:

EG_y is the net electricity supplied by the project,

$EG_{output,y}$ is the electricity supplied by the proposed project and the Nanguo wind farm;

$EG_{Nanguooutput,y}$ = Electricity generated by Nanguo wind farm

$EG_{import,y}$ is the electricity purchased from the Grid by the project and the Nanguo wind farm.

Of which, $EG_{Nanguooutput,y} = \sum_{i=1}^8 EG_{M2,i}$

Where,

$EG_{M2i,y}$ is the electricity supplied by turbine i of Nanguo wind farm measured by Meter $M2.i$;

i = The number of turbine of Nanguo wind farm, from 1 to 8.

Data in the monthly reading records were used to the report, through a cross check with ETNs, the most conservative values were used for the monitoring report and ERs calculation spreadsheet, which has been verified by the verification team. Supporting references and data required to determine the quantity of net electricity generation supplied by the project is found to be complete and transparent.

Monitoring equipment and calibration

The documents review was carried out. Particular attention was paid to the frequency of measurements, the quality of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the registered PDD and the methodology ACM0002(version 06) and corresponding tool(s), and the quality assurance and quality control procedures.

All monitoring meters have been calibrated annually as per the approved revised monitoring plan covering this monitoring period. The relevant information of the calibration is listed as below:

Monitoring equipment	Type	Serial NO.	Accuracy	Calibration frequency	Calibration date	Validity date	Calibration entity
M1	ZMD405	85618867	0.5S	Annually	10/10/2011	09/10/2012	Guangdong Power Grid Co., Ltd. Shantou Power Supply Bureau
					08/10/2012	07/10/2013	
					07/10/2013	06/10/2014	
					06/10/2014	05/10/2015	
					03/10/2015	02/10/2016	
					29/09/2016	28/09/2017	
M3	ZMD405	85618868	0.5S		26/09/2017	25/09/2018	
M2,1	DTSD1088	81101167	0.2S		06/01/2012	05/01/2013	Guangzhou Branch, National Institute of Measurement and Testing Technology
M2,2	DTSD1088	81101171	0.2S		04/01/2013	03/01/2014	
M2,3	DTSD1088	81101165	0.2S		03/01/2014	02/01/2015	
M2,4	DTSD1088	81101164	0.2S		02/01/2015	01/01/2016	
M2,5	DTSD1088	81101168	0.2S		30/12/2015	29/12/2016	
M2,6	DTSD1088	81101169	0.2S		26/12/2016	25/12/2017	
					23/12/2017	22/12/2018	

M2,7	DTSD1088	81101170	0.2S				
M2,8	DTSD1088	81101166	0.2S				
M4,1	DTSD1088	91102451	0.5S				
M4,2	DTSD1088	91102455	0.5S				
M4,3	DTSD1088	91102453	0.5S				
M4,4	DTSD1088	91102450	0.5S				
M4,5	DTSD1088	91102448	0.5S				
M4,6	DTSD1088	91102454	0.5S				
M4,7	DTSD1088	91102449	0.5S				
M4,8	DTSD1088	91102452	0.5S				

Calibration records and accreditation certificates/11//12/ have been verified by the verification team. The actual accuracy for M1, M3, M4,1-M4,8 is verified to be 0.5S and the accuracy for M2,1-M2,8 is verified to be 0.2S which are in line with the requirement in the approved revised monitoring plan. The accuracy of all meters is in compliance with the Technical administrative code of electric energy metering /15/. The calibration frequency of all meters is annual, which is consistent with the monitoring plan. Based on the site visit and by checking the calibration reports, CTI confirms that:

- All meters have been installed in accordance with the approved revised monitoring plan;
- The meters' accuracy is in line with the requirement of the approved revised monitoring plan;
- The calibration frequency of the meters is annual, which is in line with the approved revised monitoring plan and national technical standard/15/ as required by VCS standard. And the calibrations are verified to be valid for the whole reporting period.

CL01 was raised and successfully closed, please refer to Appendix D for details.

Data management and control

China Resources Wind Power (Shantou) Co., Ltd. is responsible for operation and routine maintenance of power plant under the project activity. The quality assurance and quality control procedures have been addressed in the project management and monitoring manual/5/, including the organization structure with the responsibilities, personnel competencies, monitoring procedures and monitoring management. By interview with the staff/45/ and check records /7//8/ during on-site visit, it can be confirmed that the monitoring management system is implemented following the project management and monitoring manual.

All monitoring devices have been calibrated and maintained periodically to ensure the accuracy of measurement. Calibration records of instruments used in measurements were made available during the verification visit and found to be valid for the entire period of the verification. Competence and training records of in-plant personnel engaged in measurement of plant parameters were presented during verification and found to be in order.

As the above and document review, verification team considers that the project activity operated properly in the way complying with the implementation plan and approved revised monitoring plan and the monitoring was performed in a transparent and comprehensive manner during this monitoring period.

Emission reduction calculation

CTI confirms that appropriate methods and formulae for calculating baseline emissions, project emissions and leakage have been followed, and the assumptions, emission factors and default values that are applied in the calculation have been justified.

According to the applied methodology, the emission reductions are determined as the difference between the baseline emissions, project emissions and leakage:

$$ER_y = BE_y - PE_y - L_y$$

Baseline emissions

Baseline emissions are determined as multiplying quantity of net electricity supplied to the Grid by the project (EG_y) by the validated ex-ante fixed grid emission factor (EF_y).

$$BE_y = EG_y \times EF_y$$

Combined margin CO₂ emission factor (EF_y)

$EF_{grid,CM,y}$ is the grid emission factor of the which has been verified ex-ante in the validation stage in the PDD as 0.9276 tCO₂e/MWh.

Net electricity supplied to the Grid by the project (EG_y)

As per the Section 4.1 above, EG_y is calculated as:

$$EG_y = EG_{output,y} - EG_{Nanguooutput,y} - EG_{import,y}$$

The verification team has checked the monthly reading records and ETNs issued by the grid company and emission reduction calculation sheet, and confirmed the data and calculation applied are correct, complete and transparent. The monthly electricity data and EG_y calculation are summarized in the Tables as follow:

Table 1 Monitored monthly values of EG_{output,y}, EG_{import,y} and EG_{Nanguooutput,y}

Period		EG _{output,y} from meter readings of M1	EG _{output,y} from ETNs	Conservative values of EG _{output,y}	EG _{Nanguooutput,y} measured by M2,1- M2,8	EG _{import,y} from meter readings of M1	EG _{import,y} from ETNs	Conservative values of EG _{import,y}	EG _y = EG _{output,y} -EG _{Nanguooutput,y} -EG _{import,y}
Start	End	MWh	MWh	MWh	MWh	MWh	MWh	MWh	MWh
29/09/2012	28/10/2012	5,724.400	5,682.910	5,682.910	1,324.800	30.800	30.800	30.800	4327.310
29/10/2012	30/11/2012	8,342.400	8,283.592	8,283.592	1,720.680	4.400	4.400	4.400	6558.512
01/12/2012	31/12/2012	10,516.000	10,438.939	10,438.939	1,840.320	4.400	4.400	4.400	8594.219
Subtotal 2012		-	-	24,405.441	4,885.800	-	-	39.600	19,480.041
01/01/2013	28/01/2013	8,663.600	8,596.104	8,596.104	1,610.520	8.800	8.800	8.800	6976.784
29/01/2013	26/02/2013	8,100.400	8,032.517	8,032.517	1,298.040	4.400	4.400	4.400	6730.077
27/02/2013	28/03/2013	5,183.200	5,143.519	5,143.519	1,017.120	13.200	13.200	13.200	4113.199
29/03/2013	27/04/2013	6,978.400	6,920.604	6,920.604	1,565.400	17.600	17.600	17.600	5337.604
28/04/2013	28/05/2013	4,747.600	4,712.207	4,712.207	981.000	13.200	13.200	13.200	3718.007
29/05/2013	27/06/2013	3,462.800	3,434.394	3,434.394	722.880	17.600	17.600	17.600	2693.914
28/06/2013	28/07/2013	2,178.000	2,159.803	2,159.803	381.960	17.600	17.600	17.600	1760.243
29/07/2013	28/08/2013	2,094.400	2,079.282	2,079.282	363.240	26.400	26.400	26.400	1689.642
29/08/2013	28/09/2013	6,833.200	6,781.371	6,781.371	1,695.960	35.200	35.200	35.200	5050.211
29/09/2013	28/10/2013	4,989.600	4,950.161	4,950.161	1,018.800	8.800	8.800	8.800	3922.561
29/10/2013	28/11/2013	10,986.800	10,906.657	10,906.657	2,650.920	4.400	4.400	4.400	8251.337
29/11/2013	28/12/2013	9,468.800	9,402.090	9,402.090	2,112.360	4.400	4.400	4.400	7285.330
Subtotal 2013		-	-	73,118.709	15,418.200	-	-	171.600	57,528.909
29/12/2013	25/01/2014	6,894.800	6,846.175	6,846.175	1,112.520	4.400	4.400	4.400	5729.255
26/01/2014	28/02/2014	6,991.600	6,921.804	6,921.804	1,503.000	8.800	8.800	8.800	5410.004
01/03/2014	28/03/2014	6,529.600	6,456.314	6,456.314	1,107.600	17.600	17.600	17.600	5331.114
29/03/2014	28/04/2014	4,589.200	4,547.780	4,547.780	717.240	17.600	17.600	17.600	3812.940
29/04/2014	28/05/2014	4,483.600	4,436.411	4,436.411	870.240	17.600	17.600	17.600	3548.571
29/05/2014	28/06/2014	5,042.400	4,970.242	4,970.242	851.400	17.600	17.600	17.600	4101.242
29/06/2014	28/07/2014	1,804.000	1,783.773	1,783.773	455.640	17.600	17.600	17.600	1310.533
29/07/2014	28/08/2014	1,592.800	1,574.411	1,574.411	297.000	30.800	30.800	30.800	1246.611
29/08/2014	28/09/2014	1,848.000	1,828.393	1,828.393	363.120	39.600	39.600	39.600	1425.673

29/09/2014	28/10/2014	6,674.800	6,597.496	6,597.496	1,116.240	35.200	35.200	35.200	5446.056
29/10/2014	28/11/2014	8,016.800	7,925.614	7,925.614	1,297.680	13.200	13.200	13.200	6614.734
29/11/2014	31/12/2014	10,472.000	10,350.411	10,350.411	1,921.560	4.400	4.400	4.400	8424.451
Subtotal 2014		-	-	64,238.824	11,613.240	-	-	224.400	52,401.184
01/01/2015	28/01/2015	9,512.800	9,390.035	9,390.035	1,976.280	4.400	4.400	4.400	7409.355
29/01/2015	26/02/2015	6,490.000	6,424.003	6,424.003	1,196.880	8.800	8.800	8.800	5218.323
27/02/2015	28/03/2015	7,656.000	7,578.747	7,578.747	1,659.600	13.200	13.200	13.200	5905.947
29/03/2015	28/04/2015	5,460.400	5,398.731	5,398.731	882.360	22.000	22.000	22.000	4494.371
29/04/2015	28/05/2015	3,669.600	3,620.980	3,620.980	847.800	22.000	22.000	22.000	2751.180
29/05/2015	28/06/2015	3,022.800	2,989.352	2,989.352	567.240	13.200	13.200	13.200	2408.912
29/06/2015	28/07/2015	4,382.400	4,336.217	4,336.217	846.240	13.200	13.200	13.200	3476.777
29/07/2015	28/08/2015	1,927.200	1,901.555	1,901.555	375.840	30.800	30.800	30.800	1494.915
29/08/2015	28/09/2015	3,190.000	3,158.615	3,158.615	807.480	35.200	35.200	35.200	2315.935
29/09/2015	29/10/2015	6,498.800	6,420.926	6,420.926	1,064.520	30.800	30.800	30.800	5325.606
30/10/2015	29/11/2015	7,902.400	7,817.666	7,817.666	1,292.280	8.800	8.800	8.800	6516.586
30/11/2015	31/12/2015	8,452.400	8,367.762	8,367.762	1,924.800	13.200	13.200	13.200	6429.762
Subtotal 2015		-	-	67,404.589	13,441.320	-	-	215.600	53,747.669
01/01/2016	29/01/2016	8,307.200	8,222.952	8,222.952	2,011.440	13.200	13.200	13.200	6198.312
30/01/2016	27/02/2016	6,912.400	6,828.152	6,828.152	1,140.960	4.400	4.400	4.400	5682.792
28/02/2016	29/03/2016	8,038.800	7,954.552	7,954.552	1,277.280	8.800	8.800	8.800	6668.472
30/03/2016	28/04/2016	2,481.600	2,397.352	2,397.352	404.520	13.200	13.200	13.200	1979.632
29/04/2016	30/05/2016	4,936.800	4,852.552	4,852.552	980.280	26.400	26.400	26.400	3845.872
31/05/2016	29/06/2016	2,864.400	2,780.152	2,780.152	606.000	17.600	17.600	17.600	2156.552
30/06/2016	29/07/2016	1,883.200	1,798.952	1,798.952	459.600	26.400	26.400	26.400	1312.952
30/07/2016	30/08/2016	2,569.600	2,485.352	2,485.352	643.560	26.400	26.400	26.400	1815.392
31/08/2016	29/09/2016	4,228.400	4,228.400	4,228.400	963.120	44.000	44.000	44.000	3221.280
30/09/2016	30/10/2016	7,722.000	7,722.000	7,722.000	1,202.640	26.400	26.400	26.400	6492.960
31/10/2016	28/11/2016	7,920.000	7,840.800	7,840.800	1,611.720	17.600	17.600	17.600	6211.480
29/11/2016	28/12/2016	9,944.000	9,844.560	9,844.560	2,307.600	12.760	12.760	12.760	7524.200
Subtotal 2016		-	-	66,955.776	13,608.720	-	-	237.160	53,109.896
29/12/2016	24/01/2017	10,401.600	10,297.584	10,297.584	2,108.400	4.840	4.840	4.840	8184.344
25/01/2017	26/02/2017	9,640.400	9,543.996	9,543.996	1,636.320	4.400	4.400	4.400	7903.276

27/02/2017	29/03/2017	7,937.600	7,858.224	7,858.224	1,395.840	13.200	13.200	13.200	6449.184
30/03/2017	28/04/2017	4,690.400	4,643.496	4,643.496	910.560	8.800	8.800	8.800	3724.136
29/04/2017	29/05/2017	3,898.400	3,859.416	3,859.416	814.320	26.400	26.400	26.400	3018.696
30/05/2017	28/06/2017	3,942.400	3,902.976	3,902.976	656.040	22.000	22.000	22.000	3224.936
29/06/2017	29/07/2017	1,161.600	1,149.984	1,149.984	222.600	22.000	22.000	22.000	905.384
30/07/2017	29/08/2017	3,137.200	3,105.828	3,105.828	656.640	39.600	39.600	39.600	2409.588
30/08/2017	28/09/2017	2,578.400	2,552.616	2,552.616	464.880	26.400	26.400	26.400	2061.336
29/09/2017	30/10/2017	7,876.000	7,797.240	7,797.240	1,829.640	39.600	39.600	39.600	5928.000
31/10/2017	28/11/2017	11,127.600	11,016.324	11,016.324	2,052.960	8.800	8.800	8.800	8954.564
31/10/2017	29/12/2017	9,653.600	9,557.064	9,557.064	1,584.720	4.400	4.400	4.400	7967.944
30/12/2017	31/12/2017	0.000	0.000	0.000	0.000	4.400	4.400	4.400	-4.400
Subtotal 2017		-	-	75,284.748	14,332.920	-	-	224.840	60,726.988
Total		-	-	371,408.087	73,300.200	-	-	1,113.200	296,994.687

Note:

1. As the cut-off time is at 24:00 on one day from 24th to the last day of each month, monthly readings of $EG_{output,y}$ and $EG_{import,y}$ are cross-checked with the ETNs issued by grid company, the most conservative values are applied for calculation of EG_y . The monthly readings of $EG_{Nanguooutput,y}$ are checked by CTI with the original copy of the reading records during on-site visit and confirmed that the values of $EG_{Nanguooutput,y}$ used to calculate ERs are correct and complete.

2. Since the cut-off time in December for each year is not always on the last day i.e. the 31st Dec, the electricity data for 29/12/2013-31/12/2013 and 29/12/2016-31/12/2016 is settled in Jan 2014 and Jan 2017 respectively, therefore, the ERs generated for 29/12/2013-31/12/2013 and 29/12/2016-31/12/2016 are included in the vintage 2014 and 2017 respectively. CTI accepts this separation approach as it does not result in overcalculation of ERs.

3. As the electricity data is monthly recorded and the ETNs are issued monthly, there are no separate reading records nor ETNs for the last two days of this monitoring period, i.e. 30th -31st Dec 2017. To be conservative, electricity exported by project ($EG_{output,y} - EG_{Nanguooutput,y}$) for those 2 days is considered as 0 while the whole month's (30/12/2017-28/01/2018) values of $EG_{import,y}$ are applied for ERs calculations. CTI accepts this approach as it is reasonable and conservative

The corresponding baseline emissions during this monitoring period is calculated as:

Year	EG _y (MWh)	EF _y (tCO ₂ /MWh)	BE _y (tCO ₂)
2012 (29/09/2012- 31/12/2012)	19,480.041	0.9276	18,069
2013 (01/01/2013- 28/12/2013)	57,528.909	0.9276	53,363
2014 (29/12/2013- 31/12/2014)	52,401.184	0.9276	48,607
2015 (01/01/2015- 31/12/2015)	53,747.669	0.9276	49,856
2016 (01/01/2016- 28/12/2016)	53,109.896	0.9276	49,264
2017 (29/12/2016- 31/12/2017)	60,726.988	0.9276	56,330
Total	296,994.687	-	275,489

Project emissions

As statement in the registered PDD, for the wind power activities, the project emissions from the project is zero. Hence, PE_y during the monitoring period from Guangdong Nan Ao 26MW Wind Power Project is zero.

Leakages

Leakage does not need to be accounted for this project as per the monitoring methodology ACM0002 (version 06).

Emission reductions

The emission reductions for this monitoring period were calculated as:

Monitoring period	GHG emission reductions or removals (tCO ₂ e)
29/09/2012- 31/12/2012	18,069
01/01/2013- 28/12/2013	53,363
29/12/2013- 31/12/2014	48,607
01/01/2015- 31/12/2015	49,856

01/01/2016- 28/12/2016	49,264
29/12/2016- 31/12/2017	56,330
Total ERs claimed	275,489

Comparison of actual emission reductions or net anthropogenic GHG removals by sinks with estimates in registered PDD

The emission reductions claimed are 275,489 tCO₂e in this monitoring period (i.e. 1,920 days). Compared with expected emission reductions 281,030 tCO₂e (calculated as 53,425/365d*1,920d), the reported emission reductions in this monitoring period are 1.97% lower than that expected in the registered PDD during this monitoring period. Considering the natural fluctuation of the wind resource, CTI is able to confirm that the actual emission reductions reported in this monitoring period is reasonable and the additionality of the project is not impacted.

CTI verified the input data for calculating emission reductions and the calculating process, and confirmed the result were complete and transparent. Therefore, CTI is able to confirm that the actual power supply and also emission reductions reported in this monitoring period are reasonable and appropriate.

4.5 Quality of Evidence to Determine GHG Emission Reductions and Removals

The assessment of each parameter related to calculate the GHG emission reductions is justified in Sections 4.1 and 4.4 above. Via the on-site inspection and desk review, the verification team verified the supporting evidence provided by the project proponent as listed in Appendix 3 used to determine the GHG emission reductions

For each reported data, the evidence is provided and verified as sufficient and quality is appropriate. Also, the cross-checks have been performed on the reported data with different source of evidence. The information flow from data generation and aggregation, to recording, calculation and final transposition into the monitoring report has been assessed by VVB in Section 4.4. Moreover, the calibration is verified as stated in Section 4.1 via on-site inspection with evidences to confirm the meters' location, accuracy and calibration frequency are in line with requirement of the monitoring plan.

Therefore, it is concluded that the evidence provided are verified as sufficient and quality is appropriate and thus the evidence can be used to determine the GHG emission reductions and removals for this monitoring period.

4.6 Non-Permanence Risk Analysis

The project is not AFOLU project, and thus non-permanence risk analysis is not applicable for the project.

5 VERIFICATION CONCLUSION

Shenzhen CTI International Certification Co., Ltd (CTI) has performed the verification of the emission reductions that have been reported for the project activity “Guangdong Nan Ao 26MW Wind Power Project” in China (VCS Project ID: 254) for the period 29/09/2012-31/12/2017.

The verification is based on the baseline and monitoring methodology ACM0002 (version 06), the PDD. The verification consisted of the following three phases: i) desk review of the project design and the baseline and monitoring plan; ii) follow-up interviews with project stakeholders; iii) resolution of outstanding issues and the issuance of the final verification and certification report.

The project proponents are responsible for the collection, calculation and determination of the GHG data in accordance with the approved revised monitoring plan and the reporting of GHG emission reductions on the basis set out within the project monitoring report.

Our verification and validation approach was based on the requirements as defined under the applicable VCS Standard Version 4.3 and relevant UNFCCC requirements. Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. The verification can confirm that:

- the project is implemented and operated as per the registered PDD ;
the approved revised monitoring plan is as per the applied methodology;
- the monitoring complies with the approved revised monitoring plan;
- the monitoring report and other supporting documents provided are complete and verifiable and in accordance with the applicable VCS Standard Version 4.3 and CDM requirements;
- the installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately;
- the monitoring system is in place and generates GHG emission reductions data;
- the GHG emission reductions are calculated without material misstatements;
- the project deviation is appropriately described and justified; the deviation was verified by the verification team as valid, and the project remains in compliance with the VCS rules.

It is CTI’s responsibility to provide an independent verification statement on the reported GHG emission reductions for the project. Based on an understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these, CTI planned and performed our work to obtain the information and explanations that we considered necessary to provide reasonable assurance that reported GHG emission reductions are fairly stated.

CTI does not assume any responsibility towards the issuance and utilization of the VCUs hereby verified and certified. Request for issuance of VCUs shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to CTI and the engagement conditions detailed in this report. CTI cannot be held liable by any party for decisions made or not made based on this report.

In CTI's opinion the GHG emissions reductions of the "Guangdong Nan Ao 26MW Wind Power Project" for the period 29/09/2012-31/12/2017 are fairly stated in the monitoring report (version 03 dated 24/02/2023). The GHG emission reductions were calculated correctly on the basis of the approved methodology ACM0002 (version 06) and the approved revised monitoring plan.

CTI can confirm that the GHG emission reductions are calculated without material misstatements. Based on the evidence and information that are considered necessary to guarantee that GHG emission reductions are appropriately calculated, CTI confirms the following statement:

Reporting period: 29/09/2012-31/12/2017

Verified GHG emission reductions and removals in the above verification period:

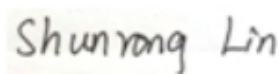
Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
2012 (29/09/2012- 31/12/2012)	18,069	0	0	18,069
2013 (01/01/2013- 28/12/2013)	53,363	0	0	53,363
2014 (29/12/2013- 31/12/2014)	48,607	0	0	48,607
2015 (01/01/2015- 31/12/2015)	49,856	0	0	49,856
2016 (01/01/2016- 28/12/2016)	49,264	0	0	49,264
2017 (29/12/2016- 31/12/2017)	56,330	0	0	56,330
Total	275,489	0	0	275,489



Mr. Feng Tian

Team Leader

27-February-2023



Ms. Lin Shunrong

Technical Reviewer

27-February-2023

APPENDIX A: ABBREVIATIONS

CAR	Corrective Action Request
CCER	Chinese Certified Emission Reduction
CER	Certified Emission Reduction(s)
CL	Clarification request
CO ₂	Carbon Dioxide
CO ₂ e	Carbon Dioxide Equivalent
CSPG	China Southern Power Grid
CTI	Shenzhen CTI International Certification Co., Ltd
DOE	Designated Operational Entity
EF	Emission Factor
EIA	Environmental Impact Assessment
ER	Emission Reduction
ETN	Electricity Transaction Note
EU ETS	EU Emissions Trading Scheme
FAR	Forward Action Request
FSR	Feasibility Study Report
GHG	Greenhouse Gas(es)
IREC	International Renewable Energy Certificates
MR	Monitoring Report
PD	Project Description
PP	Project Proponent
SN	Serial Number
VCS	Verified Carbon Standard
VCU	Verified Carbon Unit

APPENDIX B: REFERENCES

Documentation used to verify the information provided by the project proponents

/1/	Timing Carbon Asset Management Co., Ltd.: VER/VCU Monitoring Report for Guangdong Nan Ao 26MW Wind Power Project, version 01 dated 26/07/2022, version 02 dated 15/09/2022, version 03 dated 24/02/2023
/2/	Timing Carbon Asset Management Co., Ltd.: Emission reduction calculation spreadsheet for Guangdong Nan Ao 26MW Wind Power Project, version 02 dated 15/09/2022
/3/	China Resources Wind Power (Shantou) Co., Ltd. and Guangdong Power Grid Co., Ltd. Shantou Power Supply Bureau: Power Purchase Agreement for Guangdong Nan Ao 26MW Wind Power Project
/4/	China Resources Wind Power (Shantou) Co., Ltd.: Diagram of power connection system
/5/	China Resources Wind Power (Shantou) Co., Ltd.: VER monitoring manual and management procedure.
/6/	China Resources Wind Power (Shantou) Co., Ltd.: Records of training for on-site staff.
/7/	China Resources Wind Power (Shantou) Co., Ltd.: Daily operation and maintenance records.
/8/	China Resources Wind Power (Shantou) Co., Ltd.: Monthly reading records of Guangdong Nan Ao 26MW Wind Power Project from 29/09/2012-31/12/2017
/9/	Guangdong Power Grid Co., Ltd. Shantou Power Supply Bureau.: Electricity Transaction Notes from 29/09/2012-31/12/2017
/10/	China Resources Wind Power (Shantou) Co., Ltd.: Registered CDM-PDD (version 4 dated 12/03/2008) TÜV SÜD Industrie Service GmbH: CDM Validation Report, Revision No.3, dated 22/07/2008
/11/	China National Accreditation Service for Conformity Assessment: Accreditation certificates of Guangdong Power Grid Co., Ltd. Shantou Power Supply Bureau and Guangzhou Branch, National Institute of Measurement and Testing Technology which validity covering this monitoring period 29/09/2012-31/12/2017
/12/	Guangdong Power Grid Co., Ltd. Shantou Power Supply Bureau: Calibration certificates for meters: M1 and M3 covering this monitoring period Guangzhou Branch, National Institute of Measurement and Testing Technology: Calibration certificates for meters: M2,1-M2,8 and M4,1-M4,8 covering this monitoring period
/13/	Business license of China Resources Wind Power (Shantou) Co., Ltd.

/14/	China Resources Wind Power (Shantou) Co., Ltd.: Meeting minutes between the contact person and local stakeholders within this monitoring period
/15/	State Economic and Trade Commission: Technical administrative code of electric energy metering (DL/T 448-2016)
/16/	UNFCCC, http://cdm.unfccc.int
/17/	Gold Standard, http://www.goldstandard.org/
/18/	China CCER Exchange Info-platform, http://cdm.ccchina.org.cn/ccer.aspx
/19/	Measures for the Administration of Carbon Emission Trading, issued by Ministry of Ecology and Environment of China, dated 31/12/2020
/20/	Enforced company list in China cap & trade scheme, issued by Ministry of Ecology and Environment of China
/21/	China Resources Wind Power (Shantou) Co., Ltd.: Statement on the annual income of employees from 2012 to 2017
/22/	Per capita disposable income of residents in Guangdong Province in 2012 and 2017 http://stats.gd.gov.cn/
/23/	The revised monitoring plan approved by CDM EB on 09/07/2010 Validation report of revised monitoring plan
/24/	VER/VCU Monitoring Report for Guangdong Nan Ao 26MW Wind Power Project of previous monitoring period 01/01/2008-14/12/2008
/25/	TÜV Rheinland (China) Ltd.: VCS Verification Report for Guangdong Nan Ao 26MW Wind Power Project of previous monitoring period 01/01/2008-14/12/2008
/26/	CDM/CER Monitoring Reports for Guangdong Nan Ao 26MW Wind Power Project of previous CDM monitoring periods 15/12/2008-28/10/2009, 29/10/2009-28/09/2010, 29/09/2010-28/10/2011 and 29/10/2011-28/09/2012
/27/	CDM Verification Reports for Guangdong Nan Ao 26MW Wind Power Project of previous monitoring period 15/12/2008-28/10/2009, 29/10/2009-28/09/2010, 29/09/2010-28/10/2011 and 29/10/2011-28/09/2012
/28/	Goldwind Science & Technology Co., Ltd.: Nameplates of the equipment
/29/	China Resources Wind Power (Shantou) Co., Ltd.: Commissioning report
/30/	Feasibility Study Report of Guangdong Nan Ao 26MW Wind Power Project
/31/	Shantou Environmental Protection Institute: EIA Report of Guangdong Nan Ao 26MW Wind Power Project dated 21/07/2004 Shantou Environment Protection Bureau: EIA approval issued on 13/08/2004

/32/	Co-signed by the Goldwind Science & Technology Co., Ltd., and China Resources Wind Power (Shantou) Co., Ltd.: Wind Turbines Purchase Agreement
/33/	China Resources Wind Power (Shantou) Co., Ltd.: Statement on no double counting
/34/	Law of the People's Republic of China on Promotion of Employment Order of the President of the People's Republic of China
/35/	Action Plan for Carbon Dioxide Peaking Before 2030 published by the State Council on 24/10/2021
/36/	Responding to Climate Change: China's Policies and Actions released by the State Council on 10/2021
/37/	Map of the project site
/38/	Layout of the wind turbines distribution
/39/	GPS measurements of all wind turbines

Methodologies, tools and other guidance

/40/	Verified Carbon Standard: VCS Standard, Version 4.3
/41/	Verified Carbon Standard: VCS Program Guide, version 4.2
/42/	Verified Carbon Standard: VCS Sectoral Scopes http://v-c-s.org/node/448
/43/	Verified Carbon Standard: Registration and Issuance Process, version 4.2
/44/	UNFCCC EB: Approved methodology, ACM0002 , version 06

Persons interviewed

/45/	Interviewed personnel	Role	Organization
	Mr. Xiao Weinan	Plant Director	China Resources Wind Power (Shantou) Co., Ltd.
	Mr. Ke Dehao	O&M Staff	
	Mr. Zhang Donf		
	Mr. Hou Xiaohe		
	Mr. Chen Ruibiao	Local resident	Local Village
	Mr. Ke Qiushan		
	Mr. Ke Shusong		
	Mr. Lin Zeqin	Staff	Local Environment Protection Bureau
	Ms. Liu Yanyan	Project Manager	Timing Carbon Asset Management Co., Ltd.

APPENDIX C: COMPETENCE OF TEAM MEMBERS AND TECHNICAL REVIEWERS

Mr. Tian FENG

Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification					
GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
√	-	√	√	-	-

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.2: Energy generation from renewable energy sources

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by:

Wu LIN

Wu Lin

Technical Competent Manager

Shenzhen, 10/02/2022

Ms. Shunrong LIN

Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification						
Status	GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
Date	√	√	√	√	√	√

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.2: Energy generation from renewable energy sources
SS 14: Afforestation and reforestation	TA 14.1: Afforestation and reforestation
SS 15: Agriculture	TA 15.1: Agriculture

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by: *Wu Lin*

Wu LIN

Technical Competent Manager

Shenzhen, 01/01/2021

APPENDIX D: CORRECTIVE ACTION REQUESTS, CLARIFICATION REQUESTS AND FORWARD ACTION REQUESTS

Table 1: Corrective Action Requests

CAR ID	Corrective Action Request	Response by Project Proponent	Verification Team Assessment
NA	NA	NA	NA

Table 2: Clarification Requests

CL ID	Clarification Request	Response by Project Proponent	Verification Team Assessment
CL 01	MR (version 01 dated 26/07/2022) lacks the information on calibration date and validity of the electricity meters, thus CL01 was raised to clarify.	The information on calibration of the meters had been included in the updated in the MR.	OK. It has been revised and verified by CTI by checking updated MR and the calibration reports. Therefore, CL 01 is closed.

Table 3: Forward Action Requests

FAR ID	Forward Action Request	Response by Project Proponent	Verification Team Assessment
NA	NA	NA	NA